Nudge a local-AI assistance dog for deaf, blind and deafblind people

Agentverse 2026 · Robotics track · Unitree Go2 in MuJoCo · local Gemma 4 at its core · 100 % local AI

Safety always on.
Local Gemma for the unknown.

2 million+
people in the UK living with sight loss

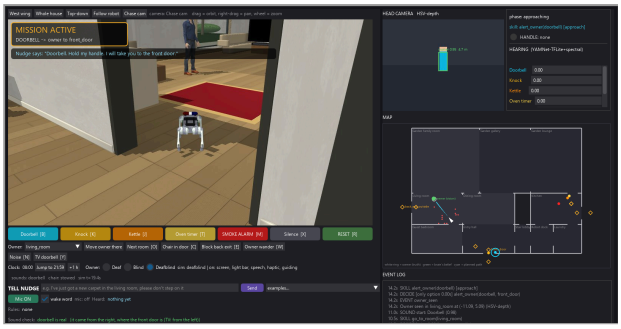
18 million
adults deaf, with hearing loss or tinnitus: 1 in 3 (RNID)

450,000+
deafblind people in the UK, rising to 610,000+ by 2035 (Sense)

16–18 months
average wait for a guide dog; some wait 2–4 years

The problem

Deafblindness is the hardest case: over 450,000 people in the UK, rising to 610,000 by 2035. An assistance dog changes their life, but there are far too few: Guide Dogs UK has only around 3,500–4,800 working partnerships, people wait 16–18 months on average (some 2–4 years), and a guide dog is reported to cost £55,000 to over £100,000 over its life. A hearing dog costs about £40,000, takes about 1.5 years to train and has a waiting list of around two years. **A deaf person who can't hear the smoke alarm can't wait two years.**



Doorbell: Nudge hears it, checks it is the real door (not the TV), finds the owner with its head camera and leads them to the front door ("Nudge says"). Perception (camera, hearing, map) shown next to the actions.

What Nudge does

Nudge is a Unitree Go2 that waits on its **charging dock** by the front door. Its stereo ears sort household sounds by urgency (**emergency**: smoke alarm; **needs action**: kettle, oven timer; **someone wants you**: doorbell, knock) and work out **where each sound comes from**. It finds the owner with its camera (or their voice), nudges them, waking them if asleep, and alerts them the way their profile needs: guide handle (deafblind), light bar and screen (deaf), speech when close (blind).

Like a real guide dog it walks on the owner's left, so its rigid **guide handle swings up on the owner's side** at hand height, where a blind owner can find it and follow. The handle gives **one pattern per urgency** (1 pulse: doorbell; 2 pulses: needs action; continuous: emergency; a glowing grip in the simulation), and the **light bar** shows which sound by colour + blinks; its red and blue ends flash only for danger. The owner taps the handle to acknowledge, then Nudge leads them to the source, or out in a fire. The owner can **talk to it** ("Nudge, take me to the kitchen") and answer its questions with a plain "yes" or "no".

The main idea

Safety is always on; everything else is learned from the owner. Smoke-alarm pre-emption, the LLM watchdog, the speed cap and calling for help are plain code that the AI or the owner can't switch off. On top, Nudge **remembers what the owner tells it**: rules ("don't step on the new carpet") and tasks ("every night at 10, take me to bed"), which Gemma turns into structured rules and code enforces and repeats.

System architecture

Two processes over ZeroMQ, so physics never waits for AI. **Simulator:** 500 Hz physics, head RGB-D camera (10 Hz), 36-ray 2D LiDAR, stereo 16 kHz ears. **Brain:** a **reflex layer** in plain code (pre-emption, obstacle stops, speed cap, watchdogs; milliseconds) and a **thinking layer**, local Gemma, woken only on events.

How we use Gemma 4 (the core)

- **Deciding from a menu.** On each event, code builds a lettered **menu of the legal options** from live perception, with facts (the chance the owner is in each room and how far, each exit's walk length and distance from the fire, unanswered nudges). Gemma answers through a **JSON schema that only allows those letters** plus a one-line reason shown on screen, so it can't invent an action. Safety decides what is on the menu; Gemma picks: which room first, re-nudge or lead out, which exit, whether to call 999.
- **Understanding the owner.** A sentence becomes a **structured rule or task** (only known rooms, objects and skills); code stores and enforces it. What it can't do ("call my daughter") is refused, never promised. Gemma also writes what Nudge says.
- **Checked every time.** A 4 s watchdog hands slow answers to rules; an exit closer to the fire than the planner's is rejected; every decision is logged, and an eval gate (90 %) re-checks Gemma after prompt changes.

On the recording laptop: **0.19 s** median per decision, **93 %** right on real logged decisions, 100 % inside the watchdog; no fine-tuning needed.

Where each model runs: all local, on one laptop with an RTX 5060, no cloud calls

Model	Job	Runs on	Latency
Gemma 4 E2B (Ollama)	**decisions + explanations; owner's words → rules / tasks**	laptop GPU	0.19 s per decision (0.26 s p95); ~0.6 s per sentence
YAMNet + spectral rules	household sounds	CPU	alert 0.5 s after a sound starts
YOLOv8n (ONNX)	owner + objects	CPU	real time at the camera's 10 Hz
Whisper base.en	owner's voice	CPU	under a second per sentence
Piper TTS	Nudge speaks	CPU	local, offline

Adapting when the scene changes

- The owner moves mid-run; a chair blocks a doorway or the back exit in a fire: the **LiDAR** sees it and the robot re-plans. When stuck it backs off, remembers the obstacle and re-routes.
- Visitors ring or knock in bursts and may give up: one visit. A doorbell from the TV is recognised as the TV (0 of 243 real doors mistaken). A kettle during a doorbell mission waits its turn. A room the camera already cleared is never re-scanned.
- **Built, off in the recorded take:** it maps a house by itself from LiDAR + depth camera (99.8 % of the floor, walls 97 % precise, exits within 0.01–0.21 m; a "Rescan house" button re-maps live); it finds its own position from LiDAR (~8 cm error vs 9.4 m drift from odometry alone); a second home (an apartment) runs with the same code.

Measured

- **Sounds:** 96 % of demo sounds detected in 1,000 random trials (5 noise profiles); smoke alarm never missed; 0 % false alerts from background noise. **Owner detection:** 97–98 % of head-camera views.
- **Adaptability:** 9 of 12 randomised, never-seen situations passed. **Teaching:** 19/19 sentences understood or refused. **Scenario tests:** default suite 14/14; robot-map mode 22/24 + 9/9 soak; apartment doorbell, smoke, kettle PASS.

Critical failure mode and how it's handled

The critical failure is **a missed or late smoke alarm, or not getting the owner out**. The alarm is checked first, in plain code, never gated: it pre-empts any task in **0.6 s** with a continuous alert. The exit is fire-aware (walk length vs distance from the fire, never through the fire room, LiDAR-blocked exits dropped), and a Gemma exit closer to the fire is rejected. Nudge asks "Shall I call 999?" while leading the owner out and calls on "yes" or no answer; if the owner isn't found within 40 s it calls anyway.

Safety

Always on, in plain code: Gemma only picks legal options; a 4 s watchdog falls back to rules; speed is capped at 0.4 m/s within 1 m of the owner; nothing the owner says can downgrade a smoke alarm; no response to repeated nudges → emergency contact, and Nudge stays.

Limits (what it cannot do yet)

- **Sound:** the demo uses synthesized clips; real recordings are weaker (smoke 96 %, doorbell 64 %).
- **The house:** room names, doors and alarm locations come from a hand-written floor plan; it only recognises furniture a standard detector knows.
- **Body:** walking is kinematic and the handle, lights and screen are simulated; self-localisation needs a known start. If the only way to the owner is blocked, it re-routes but doesn't yet escalate.
- **Next steps to a real Go2:** localisation on real sensors, real-room hearing, the handle hardware.

Sources: RNID (prevalence of deafness and hearing loss); Guide Dogs UK / Royal Canin (sight loss, partnerships); Sense (deafblindness statistics, 2023); Hearing Dogs for Deaf People; BBC, Guardian, ITV and Kent Online reporting on guide-dog waiting times and costs.

Team Nudge: Eden, Finn, Matteo, Ricktho, Sean · Scene: nerv-world house (MIT), Go2 from MuJoCo Menagerie (BSD-3) · Numbers measured in simulation on 2026-11-08.